Conditional gauge-gap transfer: multiscale control, mixing and certification

An exact retained-variable blocking scheme with hypotheses H1 and H2 below gives a volume-uniform **Wilson-transfer** energy gap along its trajectory:

$$\Delta_W(a_0) \geq \frac{\hbar c\,\gamma_m}{2^{K(a_0)}a_0}, \qquad \liminf_{a_0\downarrow 0} \Delta_W(a_0) \geq \frac{\hbar c\,\gamma_m}{a_*}, \qquad \gamma_m = \min(\gamma_*, \gamma') > 0.$$

H1 requires decay of every retained fluctuation, quasi-local coarse projections and a controlled Gibbs interaction. H2 requires uniform mixing for a whole interaction neighbourhood in every relevant periodic box. Neither hypothesis is proved for the proposed physical trajectory. A separately constructed continuum theory satisfying (E) inherits the lower bound; H1/H2 alone construct no continuum quantum field theory.

A finite-box certificate is one possible way to obtain H2 **if** an applicable finite-size theorem and its quantitative margin are supplied. It is not an automatic reduction of the problem to a small box or two coupling numbers. The exact gauge-reduction argument, robustness bound and numerical scale comparisons below state what such a certificate would need. All proof checks here are internal written checks; source reading levels are stated and no literature novelty is asserted.

Wilson measures and retained variables

Measures. The Wilson measure μ_{β_0} lives on the links of a periodic box in $(a_0\mathbb{Z})^4$ at bare coupling $\beta_0 = 6/g_0^2$. A block-spin map $\mathcal{B}$ of block size 2 is a gauge-covariant assignment of a coarse link variable to each link of the coarse lattice, measurable in the fine links, such that coarse Wilson loops are functions of fine Wilson loops. The renormalized measures $\mu^{(k)} = \mathcal{B}^k\mu_{\beta_0}$ live on the links of the periodic coarse lattices of spacing $2^k a_0$ and reproduce exactly the correlations of coarse observables. Box sides are multiples of $2^K a_0$, so that every level tiles.

Conditional expectations. Let $\mathcal{G}_k$ be the σ-algebra generated by the level-k links. Then $\mathcal{G}_0$ is generated by all fine links, and $\mathcal{G}_{k+1} \subseteq \mathcal{G}_k$ because level-$(k+1)$ links are functions of level-k links. Write $E_k = \mathbb{E}_{\mu_{\beta_0}}[\,\cdot\,|\,\mathcal{G}_k]$ and $D_k = E_k - E_{k+1}$. For a function F of level-k links, $E_{k+1}F$ is the conditional expectation given the coarse links.

Distances and supports. Distances are ℓ^∞ distances between link midpoints, measured on the torus in units of the lattice concerned. A local observable is a bounded measurable function of the links in a finite set, its support S_F. For a set S of fine links, $S^{(K)}$ is the set of level-K links within level-K distance $c_{\mathcal{B}}$ of the rescaled set $2^{-K}S$, where $c_{\mathcal{B}}$ is a constant of the block map. Fine sets at fine distance d therefore have images at level-K distance at least $d/2^K - 2c_{\mathcal{B}}$.

Interactions. An interaction $\Phi = (\Phi_X)$ assigns a bounded function of the links of X to each finite set X of links; it is gauge-invariant if every Φ_X is. Its conditional Gibbs measure on a set Λ of links, given a configuration ω outside Λ, is

$$\mu_\Lambda^{\Phi,\omega}(dU) \;\propto\; \exp\Big[-\sum_{X\cap\Lambda\neq\emptyset} \Phi_X(U\vee\omega)\Big] \prod_{\ell\in\Lambda} dU_\ell,$$

with Haar measure dU_ℓ. With range weight $\kappa > 0$ the interaction norm is $\|\Phi\|_\kappa = \sup_\ell \sum_{X\ni\ell} e^{\kappa \operatorname{diam} X}\|\Phi_X\|_\infty$, using the torus diameter on a periodic box. Total variation is $\|\mu-\nu\|_{\mathrm{TV}} = \sup_A |\mu(A)-\nu(A)|$, so that $|\mu(f)-\nu(f)| \leq 2\|f\|_\infty\|\mu-\nu\|_{\mathrm{TV}}$ for bounded f.

Multiscale and mixing hypotheses

H2 (uniform strong mixing near Φ_0). There are a gauge-invariant reference interaction Φ_0 of finite range on the coarse links, numbers $\kappa > 0$, $r > 0$, $C' < \infty$, $\gamma' > 0$ and an integer L_0 with the following property. Let T be a coarse periodic box whose sides are at least L_0, and Φ

a gauge-invariant interaction on T with $\|\Phi - \Phi_0\|_\kappa \le r$ (Φ_0 periodized). Then for every set Λ of links of T, every $W \subseteq \Lambda$, and every pair of configurations ω, ω' on $T \setminus \Lambda$ that differ only at one link y,

$$\big\| \mu_\Lambda^{\Phi,\omega}\big|_W - \mu_\Lambda^{\Phi,\omega'}\big|_W \big\|_{\mathrm{TV}} \;\le\; C'\,|W|\,e^{-\gamma'\,d(W,y)}.$$

The proof below uses this only for Λ the complement in T of a finite set.

H1 (the weak side). Along a family of bare couplings with spacings $a_0 \to 0$ (the trajectory) there are integers $K = K(a_0)$ with $2^K a_0 \to a_* \in (0,\infty)$ and a number $\gamma_* > 0$, with the following properties on every fine periodic box whose sides are multiples of $2^K a_0$ and at least $2^K L_0 a_0$ (Φ_0, κ, r, L_0 as in H2). The constants $C_k(F,G)$ and $B(F)$ may depend on a_0, k and the observables, but not on the box.

- (a) *Fluctuations.* For $k < K$ and local fine F, G at fine distance d,

$$\big|\mathbb{E}_{\mu_{\beta_0}}\big[(D_k F)(D_k G)\big]\big| \;\le\; C_k(F,G)\,e^{-\gamma_* d/2^k}.$$

- (b) *Quasi-locality of the coarse projection.* For local fine F and every integer $\ell \ge 0$ there is a function F_ℓ of the level-K links within level-K distance ℓ of $S_F^{(K)}$ with $\|E_K F - F_\ell\|_\infty \le B(F)\,e^{-\gamma_*\ell}$.

- (c) *Gibbs representation.* $\mu^{(K)}$ has density proportional to $\exp[-\sum_X \Phi_X^{(K)}]$ with respect to Haar measure on the coarse box, for a gauge-invariant interaction $\Phi^{(K)}$ with $\|\Phi^{(K)} - \Phi_0\|_\kappa \le r$.

Conditional spectral transfer

Write $\gamma_m = \min(\gamma_*, \gamma')$. A periodic box is admissible for a_0 if its sides are multiples of $2^{K(a_0)}a_0$ and at least $2^{K(a_0)}L_0 a_0$.

Theorem (conditional). Assume H1 and H2. For every a_0 on the trajectory:

1. *Coarse decay.* On the coarse lattice of every admissible box, local observables X, Y at level-K distance $D \ge 1$ satisfy

$$|\langle X;Y\rangle_{\mu^{(K)}}| \;\le\; 2C'\,|S_X|\,|S_Y|\,\|X\|_\infty\|Y\|_\infty\,e^{-\gamma' D}.$$

2. *Fine decay.* On every admissible box, local fine observables F, G at fine distance d satisfy, for every $\varepsilon \in (0, \gamma_m)$,

$$|\langle F;G\rangle_{\mu_{\beta_0}}| \;\le\; A_\varepsilon(F,G)\,e^{-(\gamma_m-\varepsilon)\,d/2^K},$$

   with $A_\varepsilon(F,G)$ finite and independent of the box.
3. *Lattice gap.* For every periodic spatial box whose side is a multiple of $2^K a_0$ and at least $2^K L_0 a_0$, the Wilson transfer matrix $\mathcal{T} = e^{-a_0 H_W/\hbar c}$ satisfies, on the gauge-invariant vectors orthogonal to its vacuum,

$$\Delta(a_0) \;\ge\; \frac{\hbar c}{a_0}\cdot\frac{\gamma_m}{2^{K(a_0)}}.$$

4. *Along the trajectory.* $\liminf_{a_0\to 0}\Delta(a_0) \ge \hbar c\,\gamma_m/a_*$, uniformly over the boxes of item 3.
5. *Continuum, conditionally.* If (E) below also holds, the continuum Hamiltonian satisfies $H \ge \hbar c\,\gamma_m/a_*$ on the vectors orthogonal to the vacuum.

(E) Continuum existence along the trajectory. There are a sequence of pairs (a_0, spatial box as in item 3) with $a_0 \to 0$, a continuum theory satisfying the Osterwalder–Schrader axioms with vacuum Ω and Hamiltonian $H \ge 0$, and a family $\mathcal{F}$ of gauge-invariant observables supported at positive Euclidean times, each F with lattice approximants F_{a_0}, such that: with ψ_{F,a_0} the lattice vector of F_{a_0} minus its vacuum component and $n = \lfloor \tau/a_0 \rfloor$, with τ a Euclidean

separation in length units (physical time τ/c), the inner products $\langle \psi_{F,a_0}, \mathcal{T}^n \psi_{F,a_0} \rangle$ converge for every $\tau \geq 0$ to $\langle \psi_F, e^{-\tau H/\hbar c} \psi_F \rangle$; and the vectors ψ_F, $F \in \mathcal{F}$, are dense in $\Omega^\perp$.

H1 and H2 alone give items 1–4. Item 5 is the only continuum statement, and its hypothesis (E) belongs to the construction problem T4 of the obligations map.

Lemma 1 (decay on a periodic box). Let Φ satisfy the bound of H2 on a coarse periodic box T, and let μ be its Gibbs measure on T. Local X, Y at distance $D \geq 1$ satisfy $|\langle X; Y \rangle_\mu| \leq 2C' |S_X| |S_Y| \, \|X\|_\infty \|Y\|_\infty e^{-\gamma' D}$.

Proof. Put $\Lambda = T \setminus S_X$. Given the configuration ω on S_X, the links of Λ have law $\mu_\Lambda^{\Phi,\omega}$, the conditional property of a finite-volume Gibbs measure. Hence $\langle X; Y \rangle = \mathbb{E}_\mu \big[X \, (\mu_\Lambda^{\Phi,\omega}(Y) - \mathbb{E}_\mu Y) \big]$, and since $\mathbb{E}_\mu Y$ is an average of $\mu_\Lambda^{\Phi,\omega'}(Y)$ over ω', $|\langle X; Y \rangle| \leq \|X\|_\infty \sup_{\omega,\omega'} |\mu_\Lambda^{\Phi,\omega}(Y) - \mu_\Lambda^{\Phi,\omega'}(Y)|$. Change ω into ω' one link of S_X at a time. Each step changes the boundary condition at one link $y \in S_X$, at distance at least D from $W = S_Y \subseteq \Lambda$, and by H2 moves $\mu_\Lambda^\Phi(Y)$ by at most $2\|Y\|_\infty C' |S_Y| e^{-\gamma' D}$. There are $|S_X|$ steps. $\square$

Lemma 2 (exact decomposition). For bounded F, G,

$$\mathrm{Cov}_{\mu_{\beta_0}}(F, G) = \mathrm{Cov}(E_K F, E_K G) + \sum_{k=0}^{K-1} \mathbb{E}\big[(D_k F)(D_k G) \big].$$

Proof. Telescoping gives $F = E_K F + \sum_{k<K} D_k F$, since $E_0 F = F$. For $j > k$, $D_j G$ is $\mathcal{G}_{k+1}$-measurable and $\mathbb{E}[D_k F \,|\, \mathcal{G}_{k+1}] = 0$, so $\mathbb{E}[(D_k F)(D_j G)] = 0$; likewise $\mathbb{E}[(D_k F)\, E_K G] = 0$ and $\mathbb{E}[D_k F] = 0$. Expand $\mathbb{E}[FG]$ and use $\mathbb{E} F = \mathbb{E}\, E_K F$. $\square$

Lemma 3 (the coarse term). Assume H1(b), H1(c) and H2. Let local fine F, G have images $S_F^{(K)}, S_G^{(K)}$ at level-K distance $D \geq 1$, and put $b_F = (1 + e^{\gamma_*})(\|F\|_\infty + B(F))$. Then

$$|\,\mathrm{Cov}(E_K F, E_K G)| \;\leq\; b_F b_G \Big[32\, C' \, |S_F^{(K)}| \, |S_G^{(K)}| \, (2D+1)^{10} + \frac{D+1}{(1 - e^{-\gamma_*})^2} \Big] e^{-\gamma_m D}.$$

Proof. Let $X_i = F_i$ from H1(b) for $i < N$ and $X_N = E_K F$, with N larger than the diameter of the coarse box; put $\Delta X_0 = X_0$ and $\Delta X_i = X_i - X_{i-1}$. Then $E_K F = \sum_{i \leq N} \Delta X_i$, $\|\Delta X_i\|_\infty \leq b_F e^{-\gamma_* i}$, and ΔX_i depends on the set A_i of level-K links within distance i of $S_F^{(K)}$. A point has at most $4(2i+1)^4$ link midpoints within ℓ^∞ distance i, so $|A_i| \leq 4(2i+1)^4 |S_F^{(K)}|$. Define ΔY_j and B_j likewise. Then $\mathrm{Cov}(E_K F, E_K G) = \sum_{i,j} \mathrm{Cov}(\Delta X_i, \Delta Y_j)$. For $i + j < D$ the sets A_i, B_j are at distance at least $D - i - j \geq 1$. By H1(c) and H2, Lemma 1 applies to $\mu^{(K)}$, and the term is at most $2C' \cdot 16 (2i+1)^4 (2j+1)^4 |S_F^{(K)}| |S_G^{(K)}| \, b_F b_G \, e^{-\gamma_*(i+j) - \gamma'(D-i-j)}$. Since $\gamma_*(i+j) + \gamma'(D-i-j) \geq \gamma_m D$, and since there are at most $(2D+1)^2$ such pairs, each with $i, j < D$, these terms sum to at most the first bracket. For $i + j \geq D$ use $|\,\mathrm{Cov}(U, V)| \leq \|U\|_\infty \|V\|_\infty$: the $s+1$ pairs with $i + j = s$ give at most $b_F b_G \sum_{s \geq D} (s+1) e^{-\gamma_* s} \leq b_F b_G (D+1) e^{-\gamma_* D} / (1 - e^{-\gamma_*})^2$. $\square$

Proof of the theorem. *Item 1* is Lemma 1 applied to $\Phi^{(K)}$, which H1(c) places in the r-ball of H2.

Item 2. By Lemma 2 the covariance is the coarse term plus the fluctuation terms. By H1(a), and since $2^k \leq 2^{K-1}$ for $k < K$, the fluctuation terms total at most $\big(\sum_{k<K} C_k(F, G) \big) e^{-2\gamma_* d / 2^K}$. The images are at level-K distance $D \geq d/2^K - 2c_{\mathcal{B}}$. For $D \geq 1$, Lemma 3 bounds the coarse term by a polynomial in D times $e^{-\gamma_m D}$, which is at most $c_\varepsilon(F, G)\, e^{-(\gamma_m - \varepsilon) D}$ with $c_\varepsilon(F, G)$ finite; for $D < 1$ use $|\,\mathrm{Cov}\,| \leq \|F\|_\infty \|G\|_\infty$. Both bounds become $e^{-(\gamma_m - \varepsilon) d / 2^K}$ times a finite constant. The constant $A_\varepsilon(F, G)$ so obtained may grow with $K(a_0)$; it does not depend on the box.

Item 3. Fix an admissible spatial box and normalize $\mathcal{T}$ so that its top eigenvalue on gauge-invariant vectors is 1, with eigenvector Ω. $\mathcal{T}$ is self-adjoint and positive (Lüscher, Commun. Math. Phys. 54 (1977) 283; Osterwalder and Seiler, Ann. Phys. 110 (1978) 440; metadata level, as in the Wilson note §3). On gauge-invariant functions its kernel is continuous and

strictly positive on a compact configuration space, so by Jentzsch's theorem the top eigenvalue is simple and isolated and $\Omega > 0$. For a bounded gauge-invariant F of the spatial links of one time slice put $\psi = (F - \langle \Omega, F\Omega\rangle)\Omega$. On the periodic box of time extent L_t, the truncated Euclidean correlation $\langle F^*; \theta_n F\rangle_{L_t}$ converges to $\langle \psi, \mathcal{T}^n \psi\rangle$ as $L_t \to \infty$ through admissible values. Item 2, with the supports at fine distance n, bounds it by $A_\varepsilon e^{-(\gamma_m - \varepsilon)n/2^K}$ uniformly in L_t. If the spectral measure of ψ gave mass $c > 0$ to $(e^{-\lambda}, 1]$ with $\lambda < (\gamma_m - \varepsilon)/2^K$, then $\langle \psi, \mathcal{T}^n \psi\rangle \geq c\, e^{-\lambda n}$ would contradict that bound for large n. Since ε is arbitrary, the spectral measure of every such ψ lives on $[0, e^{-\gamma_m/2^K}]$. These vectors are dense in the gauge-invariant vectors orthogonal to Ω, because $\Omega > 0$ almost everywhere. Hence $\|\mathcal{T}|_{\Omega^\perp}\| \leq e^{-\gamma_m/2^K}$, and item 3 follows with $\Delta = -(\hbar c/a_0) \log \|\mathcal{T}|_{\Omega^\perp}\|$. The constant A_ε entered only through its finiteness.

Item 4. $\hbar c\, \gamma_m/(2^{K(a_0)} a_0) \to \hbar c\, \gamma_m/a_*$ by H1.

Item 5. For $F \in \mathcal{F}$ the vector ψ_{F,a_0} is orthogonal to the lattice vacuum, so item 3 and the positivity of $\mathcal{T}$ give $\langle \psi_{F,a_0}, \mathcal{T}^n \psi_{F,a_0}\rangle \leq \|\psi_{F,a_0}\|^2 e^{-n a_0 \Delta(a_0)/\hbar c}$, and $n a_0 \Delta(a_0)/\hbar c \geq \lfloor \tau/a_0 \rfloor a_0 \gamma_m/(2^K a_0) \to \tau \gamma_m/a_*$. The case $\tau = 0$ of (E) gives $\|\psi_{F,a_0}\|^2 \to \|\psi_F\|^2$. Hence $\langle \psi_F, e^{-\tau H/\hbar c} \psi_F\rangle \leq \|\psi_F\|^2 e^{-\tau \gamma_m/a_*}$ for all $\tau \geq 0$. The spectral measure of H in ψ_F therefore lives on $[\hbar c \gamma_m/a_*, \infty)$, by the argument of item 3, and density of the ψ_F gives the claim. □

The constants C_k, $B(F)$ and A_ε, which may grow with $K(a_0)$, never reach item 5: the passage to the continuum needs (E) and no uniformity in those constants.

What a finite certificate would have to prove

The gauge theory is a compact-spin Gibbs specification on links, with $\Phi_p = -(\beta_W/N) \operatorname{Re} \operatorname{tr} U_p$ and $\beta_W = 2N/g^2$. Finite range and bounded plaquette terms do not by themselves supply the strong-mixing certificate required by H2. Dobrushin–Shlosman, *J. Stat. Phys.* **46** (1987) 983, and Martinelli–Olivieri, *CMP* **161** (1994) 447 and 487, are finite-size and mixing source leads at **metadata** level here. Applying their criteria to this link specification, periodic boxes and exponentially tailed generated interactions is an additional theorem, not checked merely by identifying a compact spin space.

For a finite set V of links, a perturbation with $\|\Phi - \Phi_0\|_\kappa \leq r$ changes its conditional energy by at most $|V| r$. Normalizing the two densities gives the elementary bound

$$\|\mu_V^{\Phi,\omega} - \mu_V^{\Phi_0,\omega}\|_{\mathrm{TV}} \leq \min\{1, e^{2|V|r} - 1\}.$$

Thus a **verified** quantitative finite-size condition with slack could yield a neighbourhood. This bound supplies neither the original contraction margin nor the passage from one box to all volumes. A decay-looking inequality on one finite box can be vacuous if its prefactor is allowed to be arbitrarily large.

Gauge invariance and the marginal that must be controlled

If an endpoint of a link x has all incident links in V, gauge transformations there preserve $\mu_V^{\Phi,\omega}$ for every fixed external boundary ω and act transitively by left or right multiplication on U_x. Its single-link marginal is Haar, independent of ω. A boundary-influence test using only those marginals is therefore empty. Equal one-link marginals do not imply equal joint laws. This does not invalidate the ordinary single-link Dobrushin criterion, which conditions on **all other links**, removing that free gauge transformation.

Gauge-reduction lemma. Let Φ be gauge-invariant, Λ a set of links, and G_Λ the group of gauge transformations at the sites all of whose incident links lie in Λ. For configurations ω, ω' outside Λ and $W \subseteq \Lambda$, the total-variation distance between the marginals of $\mu_\Lambda^{\Phi,\omega}$ and $\mu_\Lambda^{\Phi,\omega'}$ on W equals the supremum of $|\mu_\Lambda^{\Phi,\omega}(A) - \mu_\Lambda^{\Phi,\omega'}(A)|$ over G_Λ-invariant events A of the links of W.

Proof. An element of G_Λ moves only links of Λ, leaves ω fixed, preserves Haar measure and every Φ_X, and so preserves $\mu_\Lambda^{\Phi,\omega}$; it maps configurations of W to configurations of W. For $|f| \le 1$ on W, the Haar average Pf of f over G_Λ is invariant, satisfies $|Pf| \le 1$, and has the same expectation as f under both measures. Since the total-variation distance is half the supremum of $|\mu(f) - \nu(f)|$ over $|f| \le 1$, the supremum may be taken over invariant f. □

The invariant σ-algebra contains the traces of closed loops in W and the matrix entries of open paths in W whose endpoints are sites with an incident link outside Λ. A selection of small Wilson loops is not known to generate it. So H2 is to be read on this whole σ-algebra, boundary-attached lines included; interior single-link marginals are Haar and contribute nothing as shown by the single-link argument above.

The target is a neighbourhood of interactions

A usable target is a certified mixing set in the full normed interaction space. A blocked law has higher and longer-range terms; one scalar Wilson coupling does not describe its distance from that set. Nor is the set of gapped couplings proved to be an upper interval. Certification at one g^2 consequently certifies neither every larger coupling nor every subsequent blocked image. A schematic inequality $g^2_{\mathrm{RG}} \ge g^2_{\mathrm{DS}}$ omits both these qualifications and the observable/projection bounds H1 needs for transfer.

Even the Gibbs representation in H1(c) is substantive. A measurable block map always has a pushforward measure; it need not have a summable Gibbs interaction. Van Enter–Fernández–Sokal, hep-lat/9210032, abstract read, supplies non-Gibbsian examples, not a quantitative theorem for our map. Preserving a Gibbs representation would still leave the generated interaction norm and renewed mixing estimates to prove.

The simple oscillation method shows one failed improvement route. Using the one-link density-oscillation estimate in the Dobrushin proof, replace every block influence $\rho_V(x,y)$ by $e^{4\beta_W} - 1$. Its block sum is then bounded by $|V||\partial V|(e^{4\beta_W} - 1)$; the corresponding test $|\partial V|(e^{4\beta_W} - 1) < 1$ gets worse with the boundary. This crude majorant captures no screening. Improvement requires actual joint sub-block influence estimates, not a declaration that larger blocks are better. Its failure excludes neither other criteria nor other proofs of a gap.

Numerical orientation and its limits

The following are scale comparisons, **not certificates**. For pure $SU(3)$ use Necco–Sommer, *Nucl. Phys. B* **622** (2002) 328, metadata, for r_0/a; Morningstar–Peardon, hep-lat/9901004, abstract read, for the retained continuum-scaled input $r_0\mu_{0^{++}} = 4.21$. The orienting string-tension input is $r_0^2\sigma \simeq 1.35$. Here μ is inverse length and physical energy is $\hbar c\mu$; physical string tension is $\hbar c\sigma$.

$$\mu_{0^{++}}a = \frac{4.21}{r_0/a}, \qquad \frac{\xi}{a} = \frac{r_0/a}{4.21}, \qquad \sigma a^2 = \frac{1.35}{(r_0/a)^2}.$$

β_W	g^2	r_0/a	scaled μa	scaled ξ/a	scaled σa^2
5.7	1.05	2.92	1.44	0.70	0.16
6.0	1.00	5.37	0.78	1.28	0.047
6.2	0.97	7.38	0.57	1.75	0.025
6.4	0.94	9.74	0.43	2.31	0.014

These are continuum-scaled estimates, not measured lattice masses or uniform boundary-mixing rates. Near $\beta_W \simeq 5.7$, scalar softening beside the fundamental–adjoint first-order endpoint can increase the actual lattice correlation length; the source scope is in the crossover

comparison. Adding a negative adjoint term changes the interaction and requires its own certificate. Neither an area coefficient nor a vacuum correlation length is a boundary-influence contraction margin.

With the conventional $r_0 = 0.5$ fm, $\mu_{0^{++}} = 8.42\,\mathrm{fm}^{-1}$ and $E_{0^{++}} \simeq 1.66$ GeV using $\hbar c \simeq 0.1973$ GeV fm. The spacings are about 0.17 fm at $\beta_W = 5.7$ and 0.093 fm at 6.0. Identifying the theorem's a_* with one of these spacings is heuristic. If it were 0.093 fm and a rate $\gamma_m = 0.1$ were proved, its lower bound would be about 0.21 GeV, not the central glueball estimate. The theorem needs a positive rate, not that value.

Trial boxes and the costs not removed by symmetry

A free-boundary four-dimensional cube with R sites per axis and only internal links integrated has

$$|V| = 4(R-1)R^3 = 216, 768, 2000 \quad (R = 3, 4, 5).$$

Without gauge reduction these are $8|V| = 1728, 6144, 16000$ real integration dimensions, plus the supremum over external boundary links meeting the interaction. The counts are exact for that convention; $R = 3$–5 are trial sizes, not sufficient sizes. Proposed sizes 10–30 are likewise not a proved requirement. No certificate at the physical crossover is supplied here.

For contrast, the sufficient **fixed-cutoff Wilson** one-link condition is $\beta_W < \frac{1}{4}\log(19/18)$, implied by $g^2 > 444$. It yields

$$\Delta_W \geq \frac{\hbar c}{a} \log \frac{1}{18(e^{24/g^2} - 1)} > 0.$$

The source is the Dobrushin argument; the separate polymer argument has the sufficient integer bound $g^2 \geq 1059$. Neither supplies a margin at $\beta_W \simeq 6$, and neither is the Kogut–Susskind stability theorem. The latter has an existential local threshold, not a proved 10^{101} or 10^{102} numerical threshold. The three old coupling bands are therefore orientation relative to particular sufficient methods, not phase boundaries.

Blocking counts and an area-law countertest

Pure-$SU(3)$ one-loop running has $2b_0 \log 2 \simeq 0.09657$, hence the extrapolation

$$n \simeq \frac{g_{\mathrm{UV}}^{-2} - g_{\mathrm{ref}}^{-2}}{2b_0 \log 2}.$$

g^2_{UV}	formal n as $g^{-2}_{\mathrm{ref}} \to 0$	2^n
1	10.4	1.4×10^3
1/2	20.7	1.7×10^6
1/4	41.4	2.9×10^{12}

At the same bare-coupling convention the $SU(2)$ counts are $15.5, 31.1, 62.1$. This compares perturbative coefficients, not total proof difficulty. From $g^2 = 1/2$ to 1 the $SU(3)$ formula gives about 10.4 doublings. At fixed physical ξ, a decrease of ξ/a from 10 to 1 takes $\log_2 10 \simeq 3.32$ doublings kinematically. Neither calculation proves that earlier steps are controlled or that only the last three steps remain open. The number $K(a_0)$ in H1 is unbounded as $a_0 \downarrow 0$.

Straight-link product blocking makes a coarse plaquette exactly a $2^k \times 2^k$ fine loop. Modelling its expectation by $e^{-\sigma a^2 4^k}$ at $\sigma a^2 = 0.16$ gives 0.53, 0.077, 3.6×10^{-5} for $k = 1, 2, 3$. This omits perimeter terms, prefactors and finite-loop effects. An expectation is not a polymer activity or

an interaction norm; no three-step arrival at a certified region follows. The analogous Migdal–Kadanoff fourth-power recursion, Kadanoff *Ann. Phys.* **100** (1976) 359, metadata, is heuristic here.

Continuum construction, spectral weight and consequence for STATE

H1(a) must hold for the projections of fine observables, not merely for local functions on each coarse lattice. H1(b) is essential for the coarse covariance; H1(c) and H2 control full generated interactions. The Gaussian tube and soft-constraint estimates do not establish this nonlinear package. The weak-side and certificate questions remain open in the mechanism comparison.

The theorem's gap lower bound is neither existence nor convergence of the normalized gap in T3. An exact low-energy reduction may motivate a trajectory-matching picture, but entering a mixing neighbourhood does not prove contraction of irrelevant directions or convergence of the physical spectrum. The spectral obligations retain those separate problems. Upper spectral estimates additionally need a nonzero admissible Hamiltonian insertion and infrared moments; one observed channel may miss the global edge. Four-dimensional flow smoothness supplies neither those hypotheses nor a continuum reconstruction.

The maintained output is the conditional transfer proof and the exact gauge/robustness diagnostics, with numerical comparisons explicitly scoped. No finite certificate, normalized multiscale trajectory, continuum existence theorem or universal physical $\hbar$ is derived.